

OC Core 1.3: Source-Audited Revalidation of Collapse, Residue, and Rebirth

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Abstract

This article proves a bounded post-collapse identity classification inside the audited Ontology of Continua kernel. When a continuum loses every admissible realisation, the original continuum is dead and any surviving structure is residue rather than continuing live identity. A later live continuum is counted as rebirth only in the same-embedding branch, or in a related-embedding branch with an explicit map-and-residue witness; otherwise this article withholds the rebirth claim. The contribution is both scientific and methodological. Scientifically, the article restores a lawful theorem chain from the public Core 1.2 source corpus instead of relying on optimistic summary layers. Methodologically, it shows how unresolved theorem families are excluded rather than rhetorically blended into a stronger claim. Source witnesses are placed where they directly support the bounded result.

Problem Statement

Can the surviving source-native OC Core kernel support a bounded article on collapse, residue, and rebirth under a strict source-audit discipline? The problem is not whether the entire inherited OC corpus can be restated cleanly. The real question is whether a bounded subset of source-native claims survives a rigorous independent audit strongly enough to support a standalone mathematical foundations article. That distinction matters because the inherited package mixed preserved axioms, preserved definitions, reformulated theorem families, and noncanonical removals inside one outward-facing narrative. The article validates only the source-native rows that survive the repaired canon and dependency audit, and that support the main proof. The article does not claim a generally defended axis-state lift theorem or full external readiness of the inherited OC Core theorem corpus. Accordingly, the problem is solved only if the main theorem chain can be reconstructed from source-native material that remains canon-clean after repair, if the argument exposes where the defended case ends, and if the article remains falsifiable in the precise sense that renewed audit or contrary formal analysis could overturn it.

Main Result

The main result is a bounded classification theorem about lawful post-collapse identity, not a universal completion of the inherited ontology. Its death-and-residue route is source-native. The related-embedding rebirth material is a narrower article-local strengthening of the source rebirth predicate: this article promotes only the same-embedding branch and the explicitly witnessed related-embedding subclass, not every source-admissible related-embedding branch.

Theorem A (post-collapse identity classification). Let K be a continuum in the audited OC Core kernel. If K loses every admissible realisation under its current threshold and embedding regime, then the classification is as follows: 1. the original continuum is dead; 2. any surviving post-collapse structure is residue rather than a still-live instance of K .

Proposition B (article-local rebirth control). Within this article, a later live continuum is promoted as rebirth if and only if the audited case satisfies all of the following conditions: 1. it appears while $\Omega(K) = \emptyset$ and $k(K, t) = 0$; 2. it appears either in the same-embedding branch or in a related-embedding space M'

with $M \sim_{\text{res}} M'$; 3. for the related-embedding branch, $M \sim_{\text{res}} M'$ means that the following witness data are supplied: - a named structure-preserving map $h : M \rightarrow M'$; - a residue element r ; - the condition that $h(r) \in \Omega(K') \subseteq \Omega_{\text{amb}}(M')$; - preservation of the threshold and admissibility predicates on that residue-bearing substructure; 4. it inherits at least one structural element from residue and satisfies the birth conditions for a given level.

The canonical short label for this related-embedding control path is the article-local witness branch.

The article carries Theorem A in revised form: within the surviving source-native OC Core kernel, post-collapse traces are classified as residue. When a later qualifying live continuum appears, Proposition B classifies it as rebirth of a numerically new continuum rather than the persistence of the original continuum. The article does not present Theorem A as a new primitive source theorem. It presents it as a derived classification result whose substantive force comes from the restored death theorem and the restored irreversibility corollary; Definitions 12.5 and 12.6 do not function as independent theorem sources, but they do fix the classification criteria and lawful terminology for the two non-persistence cases. The death-to-residue route is the load-bearing theorem body. Proposition B is parallel control logic: it applies only if a later qualifying live continuum appears, and it prevents that later case from being misread as persistence of the original continuum. Theorem A therefore proves the source death-and-residue route. Proposition B records the article-local related-embedding witness branch. It does not claim that every source-admissible related-embedding rebirth is covered by the map-and-witness clause; cases without that witness remain outside this article's promoted rebirth branch. The main theorem chain consists of one revised article-level theorem, three derived lemmas, three supporting axioms, and five supporting definitions reproduced in the article. The inherited theorem families that formerly underwrote broader lift or representability claims are not used here unless they survive separate revalidation. The article therefore asks the reader to evaluate one cleanly bounded source-native argument. It does not ask the reader to accept the whole inherited framework on trust, and it does not smuggle open theorem families through stylistic compression.

Definitions and Quantitative Observables

The article fixes four load-bearing notions: - **Admissible realisation**: the nonempty live-state region that keeps a continuum lawfully live under its current thresholds and embedding regime. Later uses of “admissible-state region”, “admissible state space”, and “admissible-space collapse” refer to this same live-state object and its loss unless a narrower meaning is stated. - **Death**: the loss of that admissible realisation. - **Residue**: the structural trace that can remain after death without preserving live identity. - **Rebirth**: the emergence of a later live continuum that appears in the same-embedding branch, or in the article-local related-embedding witness branch, while $\Omega(K) = \emptyset$ and $k(K, t) = 0$; the later continuum inherits at least one structural element from residue, satisfies the relevant birth conditions, and is not identical with the original continuum.

For auditability, $\Omega(K)$ denotes the admissible-state region of K , and $k(K, t)$ denotes the continuumness scalar of K at time t . The same-embedding branch is literal. The article-local related-embedding witness branch uses the relation $M \sim_{\text{res}} M'$ from Proposition B: a named structure-preserving map $h : M \rightarrow M'$ and a residue element r must preserve the residue-bearing substructure and the threshold and admissibility predicates used to test the later continuum. The related-embedding branch is admitted only when that map/witness pair is supplied; otherwise the article promotes no related-embedding rebirth claim.

“Birth conditions for a given level” means the level-specific admissibility, threshold, and coherence conditions that allow a new continuum at that level to count as live. For the present article, those clauses are an article-level synthesis from collapse/rebirth, operators, branching topology, falsifiability, the theorem roadmap, and the worked examples under the v1.2.1 source witness and Core 1.3 repair. The required conditions are: - a nonempty residue-supported candidate state set; - embedding compatibility; - threshold satisfaction; - viable cycle or operator support; - identity distinction from the dead original; - the level-specific axes required for the proposed new continuum. Proposition B is therefore not a revalidation of every related-embedding branch allowed by the broader source Definition 12.6. It is a tightened article-local subclass: cases without the same-embedding condition or the explicit map-and-residue witness remain outside the promoted rebirth branch of this article. The article also keeps a small operational vocabulary for later projections. These observables

are not used to inflate the present argument into a full universal law; they are included because they specify what later mathematical or domain-level projections would have to measure if the bounded theorem grammar is taken seriously. These observables matter because they turn the argument from an ontological slogan into a falsifiable grammar. They tell the reader what kind of counterexample would matter, what kind of terminal loss counts as death, and what kind of post-collapse persistence claim is prohibited by the article. - **contradiction_load**: Minimal mismatch between required differences and currently expressible axes and admissible coordinates. - **minimal_lift_rank**: Smallest number of new axes or auxiliary state coordinates required to restore expressivity while holding fixed the stated contradiction data for the tested route. - **collapse_threshold**: Boundary at which no admissible realisation survives under the current thresholds and embedding constraints. - **admissible_volume_or_fiber_survival_measure**: Positive-size witness of surviving admissible states or surviving exact fibres; the identifier spelling is source-fixed, and the value is zero at collapse. - **stabilization_cost**: Margin between current contradiction load and the nearest collapse or lift-trigger threshold.

Source Audit and Canon Repair

The article is source-native in its death-and-residue theorem body. The article-local related-embedding rebirth branch is a witness-controlled extension layered on top of that audited support rather than a claim that every related-embedding rebirth route in the inherited corpus has been promoted. The main theorem chain is rebuilt from the tagged Core 1.2 LaTeX repository and the canonical compiled PDF, not from later optimistic summaries. Every load-bearing row in the main argument is required to survive three linked repairs before it can carry weight: source witnessing, canonical uniqueness, and dependency exactness. This is the central methodological discipline of the article. The tagged public source snapshot is v1.2.1 at 2c61b7879ed36cd8366d87892a066082b6418ce8. For the bounded surviving kernel, the theorem-bearing draft sources are concentrated in the axiom, core-results, and collapse/rebirth modules of that public release. The PDF witness ‘Ontology of Continua — Core v1.2.0’ is the archival compiled predecessor; the v1.2.1 tag/commit locks the source files audited alongside it. The pairing is a provenance control, not a claim that the PDF itself bears a v1.2.1 label. The revised main theorem is tracked explicitly as a bounded classification theorem whose support is distributed across anchored source rows rather than disguised as a verbatim single-source theorem. Companion dossiers extend the article with the full source witness matrix, theorem-support dossier, chronology dossier, and projection atlases, but none of those companions relax the rule that the main theorem chain must be carried by the audited Core 1.2 kernel alone. When this document refers to the companion dossier package, it means those release-facing support files rather than a separate source of theorem authority. The companion-material crosswalk is:

role	release artifact
source witness matrix	Source-audit appendix; Phase 1 closed-core dossier
theorem-support dossier	Proof-machinery appendix; proof-obligation registry
chronology and canon repair	Journal-core bridge
projection and utility atlases	Unified synthesis support dossiers, monograph Appendix Q; Practical Utility and Model Comparison Atlas; Practical Utility and Model Comparison Atlas JSON
route map	Reviewer navigation matrix
supplementary figures and tables	Root law grammar; Theorem ladder and support status; Cross-domain projection map

That repair changes the scientific meaning of publication readiness. A row cannot remain theorem-admissible merely because it once appeared inside an optimistic summary layer. If a row loses unique canonical binding, depends on unresolved rebinding, or is removed from the surviving OC theory, the row is excluded from the main article argument. In the present article, that discipline removes the inherited lift-heavy theorem family from the defended case and leaves a narrower but cleaner theorem grammar. The repaired corpus is therefore best read as an explicit partition of admissible source material. Some rows survive and support the

main theorem chain. Some rows remain scientifically interesting but require separate revalidation. Others are removed from the present article. This article treats that partition as part of the scientific contribution rather than as editorial housekeeping. Representative support inventory used directly in the present bounded argument is partitioned as follows. Source-native witnesses are listed first; derived article-level rows follow separately.

- Definition 12.1 (Collapse): A continuum K collapses at time t^* if the admissible state space becomes empty, continuumness decays to zero, and every candidate configuration violates at least one existence or stability threshold.
- Definition 12.3 (Internal collapse): Internal collapse occurs when collapse is caused solely by the internal evolution operator $E = (F, G, H, Q, R, S, U)$ acting on K while the embedding space M remains structurally static over the relevant interval.
- Definition 12.4 (External collapse): External collapse occurs when the embedding space M changes so that no configuration of K remains compatible with the new constraints, even though the prior internal evolution stayed within the previously admissible region.
- Definition 12.5 (Residue): The residue of a collapsed continuum K is the set of structures in the embedding space that remain after $\Omega(K) = \emptyset$; residues may persist without preserving the original live identity.
- Definition 12.6 (Rebirth): A rebirth event occurs when, after the collapse of K , a new live continuum K' appears in a same-embedding or related-embedding space while $\Omega(K) = \emptyset$, $k(K, t) = 0$, and $\Omega(K') \neq \emptyset$; the new continuum inherits at least one structural element from residue, satisfies the birth conditions for some level, and is not identical with the original continuum.
- Axiom 3.1 (Life conditions): A continuum is live only while its admissible realisation is nonempty and its continuumness conditions remain satisfied.
- Axiom 3.2 (Death condition): A continuum dies at time t^* when $\Omega(K(t^*)) = \emptyset$; after death the operators F, G, H, Q, R, S , and U are no longer defined for that continuum.
- Axiom 3.3 (Irreversibility of death): No operator acting within the same level can restore a dead continuum; any new live continuum is considered a new entity.
- Source Theorem 3 (Death through boundary and embedding collapse): derived from the audited source-native support chain Axiom 3.2 (Death condition), Definition 12.1 (Collapse), Definition 12.3 (Internal collapse), Definition 12.4 (External collapse).
- Source Corollary 3.2 (Irreversibility of death): derived from the audited source-native support chain Axiom 3.3 (Irreversibility of death), Definition 12.6 (Rebirth).

The derived article-level rows built on top of that source-native witness set are the following:

- Lemma 1 (Collapse entails death): derived from Source Theorem 3, Axiom 3.2, and Definitions 12.1, 12.3, and 12.4.
- Lemma 2 (Post-collapse trace is residue): derived from Source Theorem 3, Axiom 3.2, and Definitions 12.1, 12.3, 12.4, and 12.5.
- Lemma 3 (Later live continuum satisfying the rebirth predicate is classified as rebirth): derived from Axiom 3.1, Source Corollary 3.2, Axiom 3.3, and Definitions 12.5 and 12.6.
- Theorem A (post-collapse identity classification): the bounded article-level classification theorem supported by the death-and-residue route.
- Proposition B (article-local rebirth control): the parallel rebirth-control branch for later qualifying continua, including the article-local related-embedding witness discipline.

Proof Architecture and Admissible Support

The positive proof architecture is organised as a ladder instead of a slogan because the article wants the reader to see exactly where the result is justified. The ladder is review-order information, not a second theorem statement. The canonical source-native support inventory appears once in the Source Audit and Canon Repair section above, and the proof section below uses that inventory without restating it as a second source list. The route is conservative: it begins with the restored death theorem and irreversibility corollary, treats residue and rebirth as semantic controls, and excludes unvalidated lift-support families from the main article.

Lemma Chain and Proof of Theorem A

The article does not treat Theorem A as an unsupported rhetorical synthesis. It is proved as a bounded source-audited classification theorem. The source inventory is fixed in the Source Audit and Canon Repair section above; this section records only the proof composition.

Proof of Lemma 1. Axiom 3.2 gives the death condition directly, and Source Theorem 3 in the restored Core 1.2 results corpus promotes that axiom into an explicit theorem: a continuum dies when its admissible realisation becomes empty. Definition 12.1 strengthens that terminal event by adding vanishing continuumness and universal threshold failure, while Definitions 12.3 and 12.4 show that the same terminal semantics holds for both internal and external routes to collapse. Therefore the empty admissible set is not a temporary excursion; it is lawful terminal loss of

the original live continuum. Proof of Lemma 2. Lemma 1 has already established that collapse ends the life of the original continuum. Source Theorem 3 therefore rules out any reading on which a surviving post-collapse structure is still the original live object. Definitions 12.1 and 12.5 then fix the lawful classification criterion for the remaining trace: it may persist structurally in the embedding space, but only as residue and not as continuing live identity. Source anchors: **OBLIGATION::LEMMA_2**, theorem roadmap, foundational consistency, and theorem atlas. Proof of Lemma 3. Axiom 3.1 fixes the live-continuum predicate used by the later case. Source Corollary 3.2 together with Axiom 3.3 excludes any same-level or identity-preserving restoration of a dead continuum. Once that prohibition is in place, any later live continuum that appears in the same-embedding branch, or in a related-embedding space under the relation $M \sim_{\text{res}} M'$ defined above, while $\Omega(K) = \emptyset$ and $k(K, t) = 0$, and that inherits at least one structural element from residue and satisfies the birth conditions of Definition 12.6, must be numerically new rather than the persistence of the original continuum. Definition 12.6 fixes the lawful classification of exactly that later live case as rebirth. Source anchors: **OBLIGATION::LEMMA_3**, worked examples, main results, and theorem roadmap. Proof of Theorem A. Lemma 1 establishes the death of the original continuum after admissible-space collapse. Lemma 2 then classifies any surviving structure as post-collapse residue rather than continuing live identity. Theorem A is thus a derived classification theorem reconstructed from the restored death theorem and the residue criterion; it is not offered as a free-standing primitive theorem beyond the surviving source corpus. Lemma 3 supports Proposition B, not the death-to-residue route of Theorem A. No excluded lift theorem is used, no hidden representability family is imported, and no stronger transition law is presumed beyond the bounded post-collapse identity grammar stated above. Proof of Proposition B. Lemma 3 supplies the source-native non-persistence rule for a later qualifying live continuum. The article-local relation $M \sim_{\text{res}} M'$ is an additional witness discipline for the related-embedding branch: it narrows, rather than enlarges, the branch by requiring an explicit map, residue element, and preservation test before the rebirth label is promoted.

Excluded support boundary. The present route does not keep a shadow theorem body in reserve. Any inherited theorem family that would be needed to defend a general contradiction-to-lift claim is outside the main article unless it survives separate revalidation with explicit provenance and exact dependency support. At the current article boundary, no theorem family remains in provisional support limbo. The inherited lift and representability families are excluded from the main article rather than presented as tacitly available support.

Source Provenance Anchors

The anchors in this section identify exact source witnesses and audit-locks; the numbered references at the end are bibliographic citations for the same public records and the external comparison literature.

- **YASHIN_OC_CORE_2025**: Primary archival witness for the exact PDF theorem corpus audited by the present article. Alexander Yashin, Ontology of Continua — Core v1.2.0, Zenodo, 2025, DOI 10.5281/zenodo.17903912, <https://zenodo.org/records/17903912>.
- **OC_CORE_PUBLIC_TAG_V121**: Public LaTeX source lock used to witness exact theorem, axiom, and definition bodies in the audited OC corpus. Alexander Yashin, ontology-of-continua-core-main, GitHub repository, tag v1.2.1, commit 2c61b7879ed36cd8366d87892a066082b6418ce8, <https://github.com/alexanderyashin/ontology-of-continua-core-main/commit/2c61b7879ed36cd8366d87892a066082b6418ce8>.

Literature Screen

The article is not written as if it were the first text ever to discuss collapse, viability, or regime change. Its literature screen identifies relevant formal background families, states what each contributes at work-level granularity, and marks why the present article asks a different source-native identity question. The screen is not a theorem-by-theorem negative exclusion of those literatures, and it should not be read as a section-by-section priority claim against them.

External comparator families

- **GUARINO_1998**: High-level formal-ontology comparison family for category discipline and identity vocabulary. Nicola Guarino, Formal Ontology and Information Systems, in N. Guarino (ed.), Formal

Ontology in Information Systems: Proceedings of FOIS 1998, IOS Press, 1998.

- GRENON_SMITH_2004: High-level formal-ontology comparison family for dynamic identity and process structure. Pierre Grenon and Barry Smith, SNAP and SPAN: Towards Dynamic Spatial Ontology, Spatial Cognition and Computation 4(1), 2004.
- LOWE_2006: High-level metaphysical comparison family for identity and persistence categories. E. J. Lowe, The Four-Category Ontology: A Metaphysical Foundation for Natural Science, Oxford University Press, 2006.
- THOM_1994: High-level transition-taxonomy comparison family for qualitative structural change. René Thom, Structural Stability and Morphogenesis: An Outline of a General Theory of Models, Westview Press, 1994.
- AUBIN_1991: Survival-under-constraints comparison family at the present work-level review granularity. Jean-Pierre Aubin, Viability Theory, Birkhäuser, 1991.
- STROGATZ_2015: High-level comparison family for bifurcation and qualitative regime-change language. Steven H. Strogatz, Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering, 2nd ed., Westview Press, 2015.
- CONLEY_1978: High-level comparison family for persistence failure and terminal invariant-set structure. Charles Conley, Isolated Invariant Sets and the Morse Index, American Mathematical Society, 1978.
- MILNOR_1985: High-level comparison family for post-transition objects and attractor change. John Milnor, On the concept of attractor, Communications in Mathematical Physics 99(2) (1985), 177–195. The comparator claims in this section cite those works inline by the labels above: formal-ontology background uses GUARINO_1998, GRENON_SMITH_2004, and LOWE_2006; transition-taxonomy background uses THOM_1994 and STROGATZ_2015; survival-under-constraints background uses AUBIN_1991; and persistence-failure background uses CONLEY_1978 and MILNOR_1985. The following rows are article-level review classifications, not page-level or theorem-level claims about the comparator works. They state how the present article uses each comparator family as background: viability theory for admissible-state survival under constraints, the Conley-Milnor cluster for persistence-failure language and post-transition objects, and Thom/Strogatz for qualitative transition language. At this granularity, the cited families function as comparison backgrounds rather than as matched source witnesses for the exact death/residue/rebirth trichotomy defended by the OC kernel. The table keeps the Conley and Milnor works together because both are used here as transition-background comparators rather than as separate source witnesses for the OC theorem chain. | comparator | formal background role | agreement | bounded difference | work-level comparison boundary | | — | — | — | — | — | | GUARINO_1998_AND_GRENON_SMITH_2004 | formal ontology of identity and change | Formal ontology provides cited background for disciplined category vocabulary, identity conditions, and the distinction between stable objects and changing processes. | The present article uses these works only as high-level identity-and-process background at work-level granularity. | used here as a family-level ontology baseline rather than as a matched theorem chain for the post-collapse identity grammar | | LOWE_2006 | metaphysical identity and persistence categories | Metaphysical category theory supplies a cited baseline for identity discipline and for distinguishing what a thing is from how it persists through change. | The comparison remains at the level of category vocabulary and persistence framing rather than theorem-by-theorem exclusion. | used here as a family-level persistence baseline rather than as a matched theorem chain for the post-collapse identity grammar | | AUBIN_1991 | admissible-state survival under constraints | Nonempty admissible sets and threshold-governed viability determine whether a structured object remains lawfully live. | The present article uses viability theory as survival-under-constraints background at work-level granularity and then turns to a different post-collapse identity question inside the OC kernel. | used here as a survival background family, but not as a matched source for the residue/rebirth classification defended by the article | | CONLEY_1978_AND_MILNOR_1985 | Conley-Milnor comparison cluster | Invariant-set and attractor theory supplies external language for terminal loss of persistence and the appearance of new dynamical objects after transition. | The comparison remains at the level of persistence-failure language and post-transition objects rather than at theorem-by-theorem exclusion. | used here as a family-level transition background rather than as a matched theorem chain for the post-collapse identity grammar | | THOM_1994_AND_STROGATZ_2015 | qualitative transition taxonomy | These frameworks provide

broad language for structural transition, bifurcation, and regime change. | The present article uses them as transition-taxonomy background at work-level granularity rather than as theorem-by-theorem comparators. | used here as transition-taxonomy background rather than as a matched theorem chain for the post-collapse identity grammar | The source witness matrix, theorem-support dossier, and release-facing companion dossier package accompany the article as supporting materials rather than as bibliography items; the concrete paths are listed in the Source Audit and Canon Repair section above. This literature screen also bounds the novelty claim. The article does not claim to supersede catastrophe theory, viability theory, invariant-set theory, or nonlinear dynamics. Its comparative claim is narrower and sharper. Catastrophe theory analyses transition classes, viability theory studies survival under constraints, and invariant-set or attractor theory studies persistence failure and post-transition objects. The present article instead makes explicit, within the OC family, a source-audited derived classification that separates death, residue, and rebirth as distinct identity statuses under one audited chain of support.

Novelty and Prior-Art Boundary

The novelty claim in this article is exactly the bounded classification formalised in Theorem A, not a broader closure theorem for the inherited OC corpus. The main proof does not claim a universal contradiction-to-lift theorem, a full representability theory, or completed universal closure. It claims only a source-audited classification theorem on death, residue, and rebirth after admissible-state loss. The novelty claim is therefore bounded in theorem scope and deliberately modest. Relative to the inherited OC corpus, the article contributes a conservative source-audit repair that separates surviving theorem support from removed or un-revalidated material and shows that the surviving source theorem and irreversibility corollary, together with the collapse, residue, and rebirth definitions, support a post-collapse non-persistence classification. Relative to broader foundational discourse, it does not claim priority for every idea involving collapse, persistence failure, or structural rebirth. Its positive contribution is narrower and cleaner: a source-audited derived classification after admissible-state loss, together with a reproducible method for rebuilding such a classification from heterogeneous theoretical corpora without silently importing unresolved premises. Within the reviewed comparator set, the viability-kernel baseline supplies a standard external criterion for whether admissible evolution still exists under constraints. The article-level contribution then addresses a narrower downstream question inside the OC kernel by making the post-emptiness identity consequences explicit and auditable. | external baseline | relevant formal conclusion | comparison boundary at the present review granularity | added article-level content in Theorem A | — | — | — | — | Aubin viability-kernel result class | work-level background for admissible evolution under constraints | used here as a survival-under-constraints background family, not as a theorem-by-theorem source for the post-collapse identity classification defended by the article | the article asks the downstream OC-kernel identity question that begins after admissible realisation is lost | | Conley-Milnor comparison cluster | work-level background for persistence failure and post-transition objects | used here as persistence-failure background, not as a theorem-by-theorem source for the post-collapse identity classification defended by the article | the article states a bounded source-audited OC classification with an exposed exclusion boundary | The table is the novelty summary for this section. It separates external trigger-level background from the article's narrower downstream identity question and carries the canonical scope boundary for the sections that follow.

Cross-Domain Interpretation and Scope Discipline

The article has implications beyond a single formal kernel, but it does not claim that those implications are already proved as a universal cross-domain theorem. The wider Core 1.3 release separately contains a Practical Utility and Model Comparison Atlas: theorem-native domain lanes are usable under explicit theorem, data-route, and falsifier discipline, while the cross-domain route-selection lane is operationally supported within its stated bounds. Those rows are release guidance, not premises in the present journal-core theorem. The supplementary cross-domain projection map is a lane index for the proved flagship kernel, Tier 1 mathematics, ESTRA / EA extensions, domain projections, and later domain families; it is not itself a proof promotion for those lanes. This distinction is important for editorial honesty. A generalist venue needs to understand why the work matters beyond one notation system, but it also needs to see that the

article does not inflate projection potential into accomplished proof. The article therefore uses cross-domain language only to identify where the collapse-residue-rebirth grammar should be tested next and what would count as a failed projection. Tier 1 mathematics remains outside the main theorem chain in the present article. In the cross-domain projection map it functions as a supporting-evidence testbed rather than as an operational route-selection lane; route selection is handled separately by the release Practical Utility and Model Comparison Atlas. Frontier projection families remain projection targets whose admissibility depends on separate theorem programs and empirical or formal validation. None of these lanes is silently promoted into proof support for the journal-core theorem. In that sense the article offers a kernel-local collapse, residue, and rebirth grammar together with a disciplined research agenda. It does not confuse the one with the other.

Reviewer Objections and Resolutions

The article includes a concrete objection-resolution layer because recurring scientific objections should be surfaced before they return as review failures. - Objection: The inherited root slogan overreaches by treating contradiction-driven lift and collapse as equally defended in the current corpus. Resolution: The article restricts the defended claim to collapse, residue, and rebirth and explicitly excludes the general lift branch from the theorem it asks the reader to accept. - Objection: Optimistic summary layers can mix surviving, removed, and unresolved rows into one apparent theorem body. Resolution: Only canon-clean and dependency-exact rows survive into the article, and excluded rows remain outside the main theorem chain. - Objection: Collapse language can become metaphorical unless it is tied to admissible-state loss and identity rules. Resolution: The article binds collapse to admissible-space emptiness, residue to post-collapse structural trace, and rebirth to a post-collapse continuum without identity preservation. - Objection: A foundations article can overclaim cross-domain closure by narrating projection targets as accomplished proof. Resolution: The usable cross-domain routing row is bounded as operational release guidance, while projection material is carried only as a bounded projection-test map, never as completed proof.

Falsifiability and Failure Modes

The article fails under its own stated criteria if any of the following conditions is met: - a positive-body source row loses canonical uniqueness or dependency exactness under renewed audit; - an admissible-space collapse is shown to occur without the residue and rebirth distinctions used in the positive theorem grammar; - a later continuum can be identified with the original continuum without violating the article's own collapse semantics; - an excluded or unvalidated theorem family is required in order for Theorem A to go through; - the projection table is found to overstate downstream closure rather than marking bounded projection tests.

Limitations and Nonclaims

- The article does not claim universal closure of the full inherited OC Core 1.2 theorem corpus.
- It does not claim that a general contradiction-driven axis-state lift theorem is already externally defended in the main proof.
- It does not claim that the projection families listed in the supplementary cross-domain projection map are already proved across mathematics, natural science, and broader domain families.
- It does not ask the reader to equate exclusion from the main proof with scientific uselessness; several excluded theorem families remain valid revalidation targets for later papers.

Evidence and Reproducibility

The PDF/tag provenance is fixed in the Source Audit and Canon Repair section above. This section records the reproduction checks that depend on that pairing. A complete reproduction therefore requires four checks. First, the source-native claim inventory must reproduce the same surviving rows. Second, the canon repair must reproduce the same unique bindings and exact dependency outcomes for those rows. Third, the positive theorem ladder must still be reconstructible without calling excluded theorem families back into service. Fourth, the figure and table assets must regenerate from the same canonical inputs with the same captions, provenance, and bounded interpretation. Because the article is intentionally conservative, reproducibility includes the possibility of reproducing failure. If a renewed audit changes the status of a support row, the

article changes with it. The same source-native basis also preserves the explicitly declared limitations already present in Core 1.2. These limitations are not editorial noise; they are part of the truth conditions under which the present bounded theorem is submitted: - Bridging from the structural language of OC to concrete datasets remains nontrivial and requires domain-specific modelling. - Quantitative models for $K_x \rightarrow K_{x+1}$ transitions remain incomplete and are only available in partial case-specific form. - Threshold taxonomy is structurally complete, but explicit functional forms still have to be specified separately for each system class. - Operational measures of expressive capacity for realistic cognitive, social, and theoretical systems are not yet fully developed. - Inter-continuum interactions are structurally represented, but quantitative theories for complex coupled continua remain largely schematic.

Supplementary Figure and Table Callouts

The companion-material crosswalk above is the canonical support-artifact map. This section gives direct links only. Supplementary Figure 1, *Root law grammar*, summarises the bounded law of collapse, residue, and rebirth and marks the lift branch as outside the present main theorem chain. Supplementary Figure 2, *Theorem ladder and support status*, records the theorem ladder and the support boundary that separates surviving support from excluded inherited theorem families. Supplementary Table 1 summarises prospective publication families and domain projections without overstating present closure. The already usable cross-domain route-selection row is carried by the Practical Utility and Model Comparison Atlas, not by the projection table. - Supplementary Figure 1: Root law grammar. - Supplementary Figure 2: Theorem ladder and support status. - Supplementary Table 1: Cross-domain projection map.

References

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